Federated Learning: Privacy Promise vs Reality

Report

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Candidate's Declaration

I hereby certify that the work which is being presented in this report entitled "Federated Learning: Privacy Promise vs Reality" in partial fulfillment of the requirements for the award of the degree of and submitted in the of the is an authentic record of my own work carried out during the period from under the supervision of , .

The matter presented in this thesis has not been submitted by me for the award of any other degree of this or any other Institute.

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Supervisor
()

Abstract

Federated Learning (FL) is a decentralized machine learning paradigm designed to address privacy concerns associated with traditional centralized data collection. Instead of transferring raw data to a central server, FL enables model training directly on local devices, sharing only model updates. This report critically examines whether federated learning truly solves the privacy problem in machine learning. While FL reduces direct data exposure and aligns with regulatory requirements, it introduces new challenges such as statistical heterogeneity, communication overhead, and vulnerability to sophisticated attacks like data poisoning, model poisoning, and inference attacks. The report explores FL architecture, optimization techniques such as FedAvg, privacy-preserving mechanisms, and the evolving threat landscape. It concludes that federated learning is not a complete privacy solution but a promising research direction with significant open challenges.

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Chapter 1

Introduction and Motivation

The rapid advancement of machine learning has been fueled by access to large-scale data. However, despite the abundance of data, modern AI systems often struggle to utilize it effectively due to privacy, legal, and structural constraints. Regulations such as GDPR and other data protection laws restrict how data can be collected, stored, and shared. As a result, traditional centralized machine learning pipelines—where data is collected from multiple sources and stored in a single location—are increasingly impractical.

Another major issue is the **data silo problem**, where data is distributed across multiple organizations or devices and cannot be easily shared. For example, hospitals, financial institutions, and personal devices each hold valuable data, but legal and ethical constraints prevent direct data pooling.

Federated Learning emerges as a solution to these problems by fundamentally changing the learning paradigm:

> *Instead of bringing data to a central model, the model is brought to the data.*

This shift enables collaborative learning while keeping data localized. However, an important question arises:

> *Does federated learning truly guarantee privacy, or does it only reduce certain risks while introducing new ones?*

This report critically examines the privacy promise of federated learning, exploring its architecture, optimization techniques, privacy-preserving mechanisms, the evolving threat landscape, and the fundamental trade-offs involved [3, 5].

Chapter 2

Federated Learning Framework and Methodology

Federated Learning is a distributed learning framework where multiple clients (devices or organizations) collaboratively train a shared global model under the coordination of a central server [3].

2.1 Basic Workflow

The FL process typically follows these steps:

1. A global model is initialized at the server.
2. The model is sent to selected client devices.
3. Each client trains the model locally using its private data.
4. Clients send updated model parameters (not raw data) back to the server.
5. The server aggregates these updates to form a new global model.
6. The process repeats over multiple rounds.

This approach ensures that raw data never leaves the client device, which is the primary privacy advantage of FL.

2.2 Types of Federated Learning

Federated learning can be categorized based on how data is distributed:

- **Horizontal FL:** Same features, different users (e.g., multiple banks with similar attributes but different customers).

- **Vertical FL:** Same users, different features (e.g., bank + e-commerce platform sharing complementary information).
- **Federated Transfer Learning:** Neither features nor users overlap significantly.

Each type introduces different technical and privacy challenges, particularly in terms of data alignment and secure computation [5].

2.3 Why Federated Learning is Important

- Enables collaboration without violating privacy laws
- Reduces risk of centralized data breaches
- Supports edge devices (e.g., smartphones, IoT systems)
- Improves personalization while maintaining global learning

Despite these advantages, FL is not inherently secure—it only changes *where* data resides, not *how* information can be inferred.

Chapter 3

Privacy Mechanisms and Optimization

Although federated learning avoids direct data sharing, privacy is not automatically guaranteed. Several mechanisms are used to enhance privacy and efficiency.

3.1 Privacy-Preserving Techniques

3.1.1 Model Aggregation

Clients send only model updates (gradients or parameters), which are aggregated at the server. This prevents direct access to raw data but does not fully eliminate information leakage [6].

3.1.2 Differential Privacy (DP)

Differential privacy introduces noise into model updates to prevent identification of individual data points. The formal guarantee is expressed as:

$$P[\mathcal{M}(x) \in S] \leq e^{\epsilon} \cdot P[\mathcal{M}(y) \in S] \tag{3.1}$$

where:

- ϵ: privacy budget
- Lower $\epsilon \rightarrow$ stronger privacy but lower accuracy

This creates a fundamental trade-off between privacy and model performance.

3.1.3 Homomorphic Encryption

Homomorphic encryption allows computation on encrypted data so that the server never sees raw updates. While highly secure, it is computationally expensive and difficult to scale.

3.2 Federated Optimization: FedAvg Algorithm

The most widely used algorithm in FL is Federated Averaging (FedAvg) [3]:

$$G^t = G^{t-1} + \frac{\eta}{n} \sum_{i=1}^{n} \left(L_i^t - G^{t-1} \right) \tag{3.2}$$

3.2.1 Explanation

- Each client computes local updates
- Updates are averaged to form a global model
- Clients with more data often have greater influence

3.2.2 Advantages

- Reduces communication rounds
- Scalable to large networks
- Efficient compared to centralized training

3.2.3 Limitations of FedAvg

- Assumes data is similar across clients (often false)
- Sensitive to malicious updates
- Struggles with heterogeneous environments

Chapter 4

Challenges, Threats, and Defenses

While FL improves privacy at a high level, it introduces new vulnerabilities and system challenges [4].

4.1 System-Level Challenges

4.1.1 Statistical Heterogeneity (Non-IID Data)

Data across clients is often highly diverse. For example, medical data from different hospitals may follow completely different distributions. This leads to poor convergence and reduced model performance.

4.1.2 System Heterogeneity

Devices differ in:

- Computational power
- Network connectivity
- Battery constraints

This results in delays, dropped updates, and inefficient training.

4.1.3 Communication Overhead

FL requires frequent communication between server and clients, which becomes costly at scale.

4.2 Threat Model

Federated systems introduce a broader attack surface:

- **Insider attackers:** Clients or server with access to internal information
- **Outsider attackers:** Intercept communication
- **Honest-but-curious:** Follow protocol but attempt to extract information
- **Malicious attackers:** Intentionally corrupt the system

4.3 Integrity Attacks

4.3.1 Data Poisoning

Attackers manipulate training data to degrade model performance.

4.3.2 Model Poisoning

Attackers directly alter model updates to influence the global model.

4.3.3 Backdoor Attacks

A particularly dangerous attack where:

- Model behaves normally on standard inputs
- Produces incorrect outputs for specific trigger patterns

These attacks are stealthy and difficult to detect.

4.4 Privacy Attacks

Even without access to raw data, attackers can infer sensitive information:

4.4.1 Gradient Inversion

Reconstructs training data from gradients [6, 1].

4.4.2 Membership Inference

Determines whether a data point was part of training.

These attacks highlight a critical insight:

Sharing gradients is not equivalent to preserving privacy.

4.5 Defense Mechanisms

Several defenses exist but none are universally effective:

- Robust aggregation (median, trimmed mean)
- Differential privacy noise addition
- Gradient compression
- Anomaly detection for malicious clients

Key Limitation: No single defense can protect against all attack types simultaneously.

Chapter 5

Trade-offs, Emerging Directions, and Conclusion

5.1 Fundamental Trade-offs

5.1.1 Privacy vs Accuracy

Stronger privacy (more noise) reduces model performance. This is an inherent tension in differential privacy-based approaches.

5.1.2 Robustness vs Flexibility

Handling heterogeneous data often conflicts with robustness to attacks.

5.1.3 Security vs Efficiency

Encryption and secure aggregation increase computational cost, creating a trade-off between the strength of security guarantees and system efficiency.

5.2 Emerging Research Directions

To address current limitations, several approaches are being explored:

5.2.1 Personalized Federated Learning

Models adapt to individual clients rather than enforcing a single global model, enabling better performance on non-IID data.

5.2.2 Multi-center Federated Learning

Groups similar clients to improve performance by reducing the effects of statistical heterogeneity.

5.2.3 Reliable Client Selection

Uses trust or reputation systems to reduce malicious influence during the aggregation process [2].

5.2.4 Advanced Privacy Techniques

Improving differential privacy and secure aggregation to provide stronger privacy guarantees with lower utility cost.

5.3 Conclusion

Federated Learning represents a significant shift in machine learning, offering a framework that aligns with modern privacy requirements. However, it does not fully solve the privacy problem. While it eliminates direct data sharing, it introduces new vulnerabilities through model updates, system heterogeneity, and adversarial behavior.

> *Federated Learning is not a complete solution—it is an evolving direction.*

Future research must focus on developing robust, scalable, and secure federated systems that balance privacy, efficiency, and accuracy.

Appendix A

Appendix

A.1 Key Terms

Federated Learning (FL) A decentralized machine learning paradigm where model training occurs on local devices without sharing raw data.

Differential Privacy (DP) A mathematical framework that provides formal privacy guarantees by adding calibrated noise to computations.

Non-IID Data Non-Independent and Identically Distributed data, where data across clients follows different distributions.

Model Aggregation The process of combining local model updates from multiple clients into a single global model.

A.2 Example FL Workflow

A typical federated learning workflow proceeds as follows:

$$\text{Client} \rightarrow \text{Local Training} \rightarrow \text{Parameter Sharing} \rightarrow \text{Server Aggregation} \rightarrow \text{Updated Model}$$

This cycle repeats over multiple communication rounds until the global model converges.

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List of Abbreviations

LAH	**L**ist **A**bbreviations **H**ere
WSF	**W**hat (it) **S**tands **F**or

Physical Constants

Speed of Light $c_0 = 2.997\,924\,58 \times 10^8\,\mathrm{m\,s^{-1}}$ (exact)

List of Symbols

a	distance	m
P	power	W (J s^{-1})
ω	angular frequency	rad

List of Mathematical Symbols

$\{\}$	Denotes empty set.
$\{\square\}$	Denotes a singleton set: $\{x\}$ denotes the set that has x as a single element.

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